

Decimals



Curriculum Ready

ACMNA: 103, 128, 129, 130, 152, 154, 156



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Decimals allow us to be more accurate with our calculations and measurements.

Because most of us have ten fingers, it is thought that this is the reason the decimal system is based around the number 10!

So we can think of decimals as being fractions with powers of 10 in the denominator.

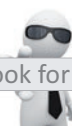
Write in this space EVERYTHING you already know about decimals.



- Q** To make dark-green coloured paint, you can mix yellow and blue together, using exactly 0.5 (half) as much yellow as you do blue.

How much dark-green paint will you make if you use **all** of the 12.5 mL of blue paint you have?

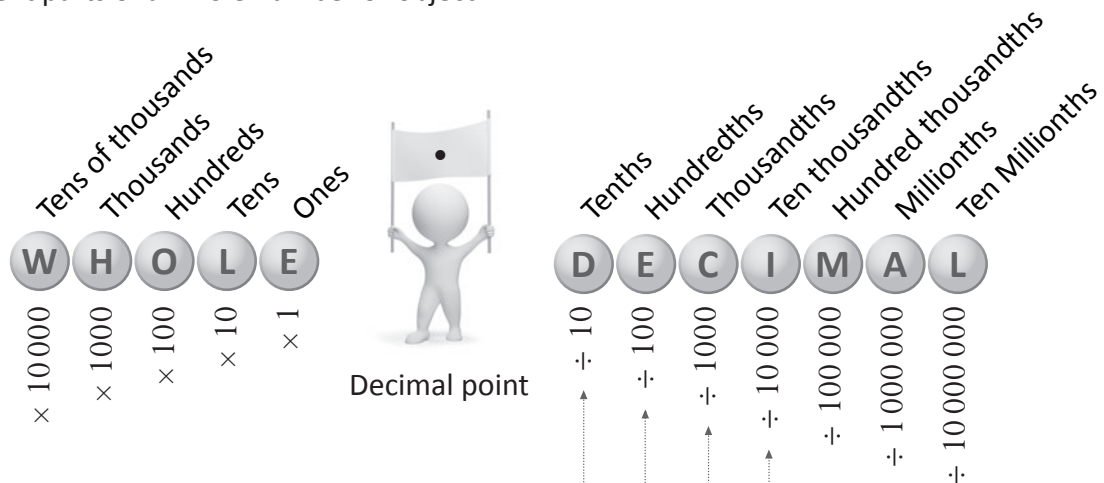
Work through the book for a great way to do this





Place value of decimals

Decimals represent parts of a whole number or object.



- 1st decimal place: $\div 10 = \times \frac{1}{10} = \text{one tenth}$
- 2nd decimal place: $\div 10 = \times \frac{1}{100} = \text{one hundredth}$
- 3rd decimal place: $\div 10 = \times \frac{1}{1000} = \text{one thousandth}$
- 4th decimal place: $\div 10 = \times \frac{1}{10000} = \text{one ten thousandth etc...}$

Add 'th' to the name for decimal place values



Write the place value of each digit in the number 465.2703

4 6 5 . 2 7 0 3

Multiply by multiples of 10

Divide by multiples of 10

Expanded forms

Place values

4.....4 × 100	= 400	= 4 hundred	Integer parts
6.....6 × 10	= 60	= 6 tens (or sixty)	
5.....5 × 1	= 5	= 5 ones (or five)	
•			
2.....2 ÷ 10 (or 2 × $\frac{1}{10}$)	= $\frac{2}{10}$	= 2 tenths	1 st decimal place
7.....7 ÷ 100 (or 7 × $\frac{1}{100}$)	= $\frac{7}{100}$	= 7 hundredths	2 nd decimal place
0.....0 ÷ 1000 (or 0 × $\frac{1}{1000}$)	= $\frac{0}{1000}$	= 0 thousandths	3 rd decimal place
3.....3 ÷ 10 000 (or 3 × $\frac{1}{10 000}$)	= $\frac{3}{10 000}$	= 3 ten thousandths	4 th decimal place



Place value of decimals

1 Write the decimal that represents these:

a 2 hundredths

0.02



Always put a zero in front
(called a **leading zero**) when
there are no whole numbers

b 9 tenths

c 1 ten thousandth

d 3 thousandths

e 6 hundred thousandths

f 8 millionths

2 Write the fraction that represents these:

a 3 tenths

b 7 thousandths

c 1 hundredth

d 9 ten thousandths

e 51 hundredths

f 11 ten thousandths

3 Write the place value of the digit written in square brackets for each of these decimals:

a [3] 1.325

b [1] 0.231

c [4] 15.046

d [5] 0.05043

e [6] 0.79264

f [0] 8.56309

4 Circle the digit found in the place value given in square brackets:

a [tenths]

8.171615

b [thousandths]

4.321230

c [hundred thousandths]

100.1001001

d [hundredths]

9.12421

e [ten thousandths]

16.123210

f [millionths]

3.120619



Place value of decimals

Each digit is multiplied by the place value and then added together when writing a number in expanded form.

Write the decimal 23.401 in expanded form

$$\begin{aligned} 23.401 &= 2 \times 10 + 3 \times 1 + 4 \times \frac{1}{10} + 0 \times \frac{1}{100} + 1 \times \frac{1}{1000} \\ &= 2 \times 10 + 3 \times 1 + 4 \times \frac{1}{10} + 1 \times \frac{1}{1000} \end{aligned}$$

Multiply each digit by its place value

Zero digits can be removed to simplify

5 Write these decimals in expanded form:

- a $4.19 =$
- b $29.281 =$
- c $40.2685 =$
- d $3.74932 =$
- e $0.2306 =$
- f $0.0085 =$

6 Simplify these numbers written in expanded form:

- a $1 \times 1 + 4 \times \frac{1}{10} + 6 \times \frac{1}{100} =$
- b $4 \times 10 + 9 \times 1 + 0 \times \frac{1}{10} + 7 \times \frac{1}{100} =$
- c $5 \times 100 + 2 \times 10 + 0 \times 1 + 2 \times \frac{1}{10} + 1 \times \frac{1}{100} + 8 \times \frac{1}{1000} =$
- d $6 \times 1 + 8 \times \frac{1}{10} + 5 \times \frac{1}{100} + 0 \times \frac{1}{1000} + 2 \times \frac{1}{10\,000} + 9 \times \frac{1}{100\,000} =$

Psst: Remember to include a **leading zero** for these ones.

- e $2 \times \frac{1}{10} + 0 \times \frac{1}{100} + 3 \times \frac{1}{1000} =$
- f $6 \times \frac{1}{100} + 7 \times \frac{1}{1000} + 0 \times \frac{1}{10\,000} + 1 \times \frac{1}{100\,000} =$
- g $3 \times \frac{1}{10} + 4 \times \frac{1}{100} + 1 \times \frac{1}{1000} + 0 \times \frac{1}{10\,000} + 8 \times \frac{1}{100\,000} =$



Approximations through rounding numbers

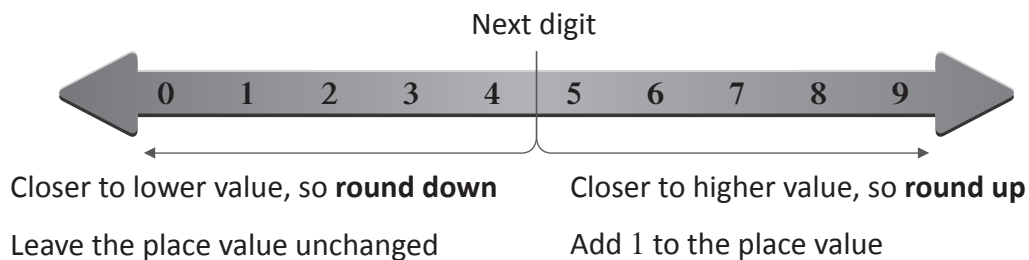


Look at these two statements made about a team of snowboarders:

- They have attempted 4937 tricks since starting = **Accurate** statement
- They have attempted nearly 5000 tricks since starting = **Rounded off** approximation

Rounding off values is used when a great deal of accuracy is not needed.

The next digit following the place value where a number is being rounded off to is the important part.



Here are some examples to see how we round off numbers.

Round these numbers

- (i) 2462 to the nearest hundred
- | | |
|----------------|--|
| 2 <u>4</u> 6 2 | The digit '4' is in the hundreds position |
| 2 4 <u>6</u> 2 | The next digit is a 6, so round up by adding 1 to 4 |
| 2 5 0 0 | Change the other smaller place value digits to 0's |

$\therefore 2462 \approx 2500$ rounded to the nearest hundred

- (ii) 0.3145 to one decimal place (or to the nearest tenth)

0 . <u>3</u> 1 4 5	The digit '3' is in the first decimal place
0 . 3 <u>1</u> 4 5	The next digit is a 1, so round down
0 . 3	Write decimal with one decimal place only

$\therefore 0.3145 \approx 0.3$ rounded to one decimal place

- (iii) 26.35819 to four decimal places (or to the nearest ten thousandth)

2 6 . 3 5 8 <u>1</u> 9	The digit '1' is in the fourth decimal place
2 6 . 3 5 8 1 <u>9</u>	The next digit is a 9, so round up by adding 1 to 1
2 6 . 3 5 8 2	Write decimal with four decimal places only

$\therefore 26.35819 \approx 26.3582$ rounded to four decimal places



Approximations through rounding numbers



1 Round these whole numbers to the place value given in square brackets.

a [nearest ten]

(i) $536 \approx$

(ii) $8514 \approx$

(iii) $93025 \approx$

b [nearest hundred]

(i) $14\,302 \approx$

(ii) $4764 \approx$

(iii) $80\,048 \approx$

c [nearest thousand]

(i) $98\,542 \approx$

(ii) $18\,401 \approx$

(iii) $120\,510 \approx$

2 Round these decimals to the decimal places given in the square brackets.

a [nearest tenth]

(i) $0.73 \approx$

(ii) $3.47 \approx$

(iii) $11.85 \approx$

b [nearest hundredth]

(i) $2.406 \approx$

(ii) $0.007 \approx$

(iii) $1.003 \approx$

c [nearest thousandth]

(i) $10.4762 \approx$

(ii) $0.3856 \approx$

(iii) $0.048640 \approx$

3 Approximate the following distance measurements:

a A group of people form an 8.82 m long line when they stand together.

(i) How long is this line to the nearest 10 cm (i.e. 1 decimal place)? $\approx$

(ii) What is the approximate length of this line to the nearest 10 metres? $\approx$

b Under a microscope the length of a dust mite was 0.000194 m

(i) Approximate the length of this dust mite to the nearest ten thousandth $\approx$ of a metre.

(ii) Approximate the length of this dust mite to the nearest hundredth of a metre. $\approx$

c If Lichen City is 3 458 532 m away from Moss City:

(i) What is this distance approximated to the nearest km? $\approx$ (i.e. nearest thousand)

(ii) What is the approximate distance between the cities to the nearest 100 km? $\approx$

(iii) Are the digits 2, 3 or even 5 important to include when describing the total distance between the two cities? Briefly explain here why/why not.





Approximations through rounding numbers

Rounding **up** can affect more than one digit when the number 9 is involved.

Round 0.95 to one decimal place

0 . 9 5

The digit '9' is in the tenths position

0 . 9 (5)

The next digit is a 5, so **round up** by adding 1 to 9

1 . 0

Change the other smaller place value digits to 0s

9 rounds up to 10, so the 9 becomes 0 and 1 is added to the digit in front.

$\therefore 0.95 \approx 1.0$ rounded to one decimal place



4 Round off these numbers according to the square brackets.

a [one decimal place]

1.98 $\approx$

b [nearest ten]

398 $\approx$

c [two decimal places]

11.899 $\approx$

d [nearest ones]

79.9 $\approx$

e [three decimal places]

0.1398 $\approx$

f [three decimal places]

2.1995 $\approx$

g [nearest thousand]

49798 $\approx$

h [nearest ones]

199.9 $\approx$

i [four decimal places]

9.89999 $\approx$

5 Approximate these values:

a A call centre receives an average of 2495.9 calls each day during one month.

(i) Approximate the number of calls received to the nearest hundreds.

$\approx$

(ii) Approximately how many thousands of calls did they receive?

$\approx$

(iii) Estimate the number of calls received daily throughout the month.

$\approx$

b A swimming pool had a slow leak, causing it to empty 9599.5896 L in one week.

(i) How much water was lost to the nearest 10 litres?

$\approx$

(ii) How much water was lost to the nearest mL if $1\text{ mL} = \frac{1}{1000}\text{ L}$?

$\approx$

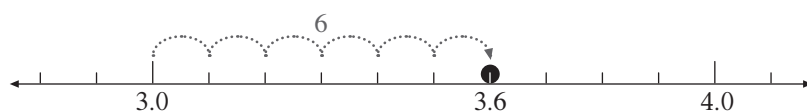
(iii) Is the digit 6 important when approximating to the nearest whole litre? Briefly explain here why/why not.



Decimals on the number line

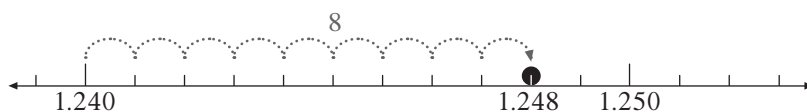
The smallest place value in a decimal is used to position points accurately on a number line.

- Decimals are based on the number 10, so there are always ten divisions between values
Eg: Here is the value 3.6 on a number line:



Six tenths of the way from 3.0 to 4.0

- The major intervals on the number line are marked according to the **second last** decimal place value

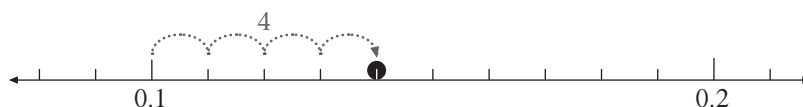


So its eight thousandths of the way from 1.240 to 1.250

Here are some more examples involving number lines:

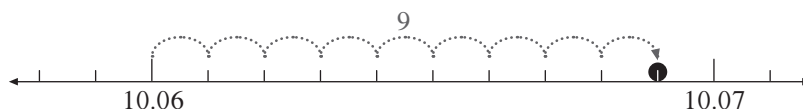
- (i) What value do the plotted points represent on the number lines below?

α)



Point is four steps from 0.1 towards 0.2, so the plotted point is: 0.14

β)



Point is nine steps from 10.06 towards 10.07, so the plotted point is: 10.069

- (ii) Round the value of the plotted points below to the nearest hundredth.

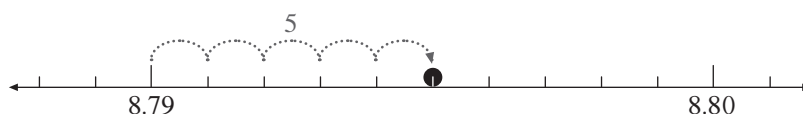
α)



Point is three steps from 2.14 towards 2.15, so the plotted point is 2.143

∴ the value of the plotted point to the nearest hundredth is: 2.14

β)

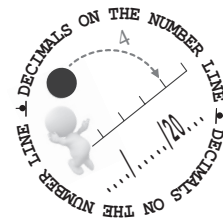


Point is five steps from 8.79 towards 8.80, so the plotted point is 8.795

∴ the value of the plotted point to the nearest hundredth is: 8.80

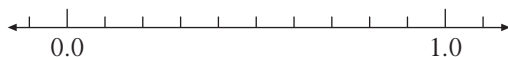


Decimals on the number line

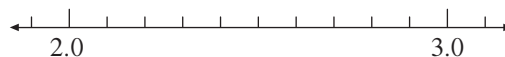


1 Display these decimals on the number lines below:

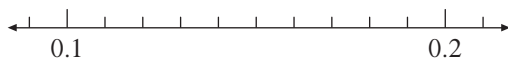
a 0.7



b 2.1



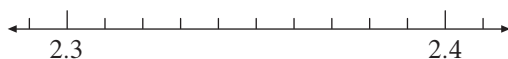
c 0.13



d 9.15



e 2.34

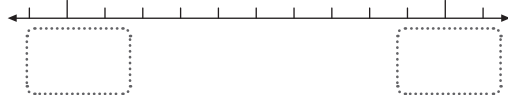


f 5.212

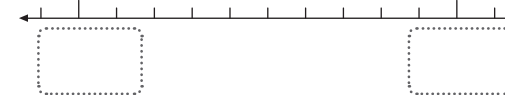


2 Label these number lines and then display the given decimal on them:

a 1.6



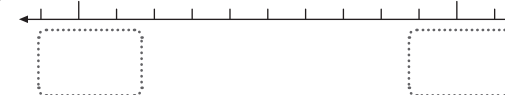
b 4.2



c 0.94



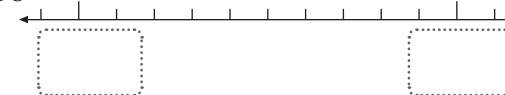
d 7.07



e 2.053

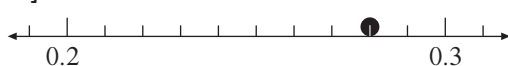


f 9.538



3 Round the value of the plotted points below to the nearest place value given in square brackets.

a [tenth]



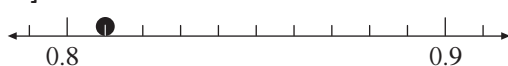
∴ the value $\approx$

b [hundredth]



∴ the value $\approx$

c [tenth]



∴ the value $\approx$

d [hundredth]



∴ the value $\approx$

e [thousandth]



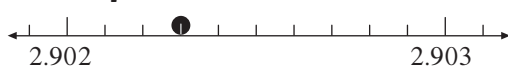
∴ the value $\approx$

f [thousandth]



∴ the value $\approx$

g [thousandth]



∴ the value $\approx$

h [thousandth]



∴ the value $\approx$

Multiplying and dividing by powers of ten

Move the decimal point depending on the number of zeros

$\otimes$ = decimal point moves right , $\div$ = decimal point moves left

Calculate these multiplication and division questions involving powers of 10:

(i) 5×1000

$$5 \times 1000 = 5.0 \times 1000 \quad \text{The whole number in decimal form}$$



We can simply add the same number of zeros to the end of the whole number

$$= 5.0 \quad \begin{array}{c} 1 \quad 2 \quad 3 \\ \curvearrowright \quad \curvearrowright \quad \curvearrowright \end{array}$$

$$= 5000$$

Fill the empty bounces with 0s

If the decimal point is on the left after dividing, an extra 0 is placed in front.

(ii) $8 \div 100$

$$8 \div 100 = 8.0 \div 100 \quad \text{The whole number in decimal form}$$

Remember to include the leading zero 😊



$$= . \begin{array}{c} 2 \quad 1 \\ \curvearrowright \quad \curvearrowright \end{array} 8.0$$

$$= 0.08$$

$\div 100$ has 2 zeros, so move decimal point 2 spaces left

Fill the empty bounces with 0s and put a zero in front

(iii) $1.25893 \times 10\,000$

$$1.25893 \times 10\,000 = 1. \begin{array}{c} 1 \quad 2 \quad 3 \quad 4 \\ \curvearrowright \quad \curvearrowright \quad \curvearrowright \quad \curvearrowright \end{array} 2589.3 \quad \text{Move decimal point 4 spaces right}$$

$$= 12\,589.3 \quad \text{No empty bounces to fill, so this is the answer}$$

(iv) $24.905 \div 100\,000$

$$24.905 \div 100\,000 = . \begin{array}{c} 5 \quad 4 \quad 3 \quad 2 \quad 1 \\ \curvearrowright \quad \curvearrowright \quad \curvearrowright \quad \curvearrowright \quad \curvearrowright \end{array} 24.905 \quad \text{Move decimal point 5 spaces left}$$

$$= 0.00024905 \quad \text{Fill empty bounces with 0s and put a zero in front}$$

(v) $260.15 \times \frac{1}{1000}$

$$260.15 \times \frac{1}{1000} = 260.15 \div 1000 \times \frac{1}{1000} \quad \text{is the same as } \div 1000$$

$$= . \begin{array}{c} 3 \quad 2 \quad 1 \\ \curvearrowright \quad \curvearrowright \quad \curvearrowright \end{array} 260.15$$

Move decimal point 3 spaces left

$$= 0.26015 \quad \text{Place a leading zero in front of the decimal point}$$

**Multiplying and dividing by powers of ten**

1 Calculate these multiplications. Remember, multiply means move decimal point to the right:

a 8×100

b 3.4×10

c 29×1000

d $12.45 \times 10\,000$

e 0.512×100

f $0.0000469 \times 1\,000\,000$

2 Calculate these divisions. Remember, divide means move decimal point to the left:

a $2 \div 100$

b $4590 \div 1000$

c $0.014 \div 10$

d $70.80 \div 10\,000$

e $1367.512 \div 1000$

f $421900 \div 100\,000\,000$

Here are some of the powers of 10 in index form. The power = the number of zeros.

$$10^1 = 10$$

$$10^2 = 100$$

$$10^3 = 1000$$

$$10^4 = 10\,000$$

$$10^5 = 100\,000$$

$$10^6 = 1\,000\,000$$

3 Calculate these mixed problems written in index form:

a 31×10^2

b $2400 \div 10^5$

c 0.0027×10^6

d 90.008×10^4

e $3.45 \div 10^3$

f $2159\,951 \div 10^7$



Multiplying and dividing by powers of ten



4

For these calculations:

(i) Show where our character needs to spray paint a new decimal point, and

(ii) write down the two numbers the new decimal point is between to solve the puzzle

a 2830.3920×100

2 8 3 0 3 9 * 2 0



I 9 and 2

b $23857 \div 1000$

2 3 8 5 7

N

c 0.4763892×10^5

0 4 7 6 3 8 9 2

A

d $382961 \div 10000$

3 8 2 9 6 2

O

e 19238.07×10^1

1 9 2 3 8 0 7

X

f 8.9236701×10000

8 9 2 3 6 7 0 1

T

g $20917983 \times \frac{1}{1000000}$

2 0 9 1 7 9 8 3

R

h $83917 \div 10^5$

8 3 9 1 7

I

i $902873.021 \times \frac{1}{10^2}$

9 0 2 8 7 3 2 0 1

D

j 0.08390×10^3

0 0 8 3 9 0

P

This is another mathematical name for a decimal point:



0 and 9 8 and 9 8 and 7 9 and 2 0 and 7 3 and 9 8 and 2 0 and 8 3 and 8 6 and 7

Terminating decimals to fractions

These have decimal parts which stop (or terminate) at a particular place value.

The place value of the **last digit on the right** helps us to write it as a fraction.



Write 0.3 as a fraction:

Decimal	Fraction
0.3	$= \frac{3}{10}$
$\uparrow$ Last digit is in tenths position	$\longleftarrow$ Decimal digits in the numerator

Integers in front of the decimal values are simply written in front of the fraction.

Write 1.07 as a fraction:

1.07	$= 1 \frac{7}{100}$	$\longleftarrow$ Decimal digits in the numerator
$\uparrow$ Last digit is in hundredths position		

07 is just 7



Always simplify the fraction parts if possible. These two examples show you how.

Write each of these decimals as an equivalent (equal) fraction in simplest form

(i) 0.25

0.25	$= \frac{25}{100}$	Equivalent, un-simplified fraction
$\uparrow$ Last digit is in hundredths position		
	$= \frac{25 \div 25}{100 \div 25}$	Divide numerator and denominator by HCF
	$= \frac{1}{4}$	Equivalent fraction in simplest form

(ii) 2.105

2.105	$= 2 \frac{105}{1000}$	Equivalent, un-simplified mixed numeral
$\uparrow$ Last digit is in thousandths position		
	$= 2 \frac{105 \div 5}{1000 \div 5}$	Divide numerator and denominator by HCF
	$= 2 \frac{21}{200}$	Equivalent mixed numeral in simplest form



Terminating decimals to fractions

1 Write each of these decimals as equivalent fractions:

a $0.1 = \frac{\boxed{}}{\boxed{}}$

b $0.7 = \frac{\boxed{}}{\boxed{}}$

c $0.09 = \frac{\boxed{}}{\boxed{}}$

d $0.03 = \frac{\boxed{}}{\boxed{}}$

e $0.001 = \frac{\boxed{}}{\boxed{}}$

f $0.007 = \frac{\boxed{}}{\boxed{}}$

g $0.013 = \frac{\boxed{}}{\boxed{}}$

h $0.049 = \frac{\boxed{}}{\boxed{}}$

i $0.129 = \frac{\boxed{}}{\boxed{}}$

j $0.081 = \frac{\boxed{}}{\boxed{}}$

k $0.1007 = \frac{\boxed{}}{\boxed{}}$

l $0.0601 = \frac{\boxed{}}{\boxed{}}$

2 Write each of these decimals as equivalent fractions and then simplify:

a $0.5 = \frac{\boxed{}}{\boxed{}} = \frac{\boxed{}}{\boxed{}}$
Simplest form

b $0.6 = \frac{\boxed{}}{\boxed{}} = \frac{\boxed{}}{\boxed{}}$
Simplest form

c $0.02 = \frac{\boxed{}}{\boxed{}} = \frac{\boxed{}}{\boxed{}}$
Simplest form

d $0.08 = \frac{\boxed{}}{\boxed{}} = \frac{\boxed{}}{\boxed{}}$
Simplest form

e $0.004 = \frac{\boxed{}}{\boxed{}} = \frac{\boxed{}}{\boxed{}}$
Simplest form

f $0.005 = \frac{\boxed{}}{\boxed{}} = \frac{\boxed{}}{\boxed{}}$
Simplest form

g $0.12 = \frac{\boxed{}}{\boxed{}} = \frac{\boxed{}}{\boxed{}}$
Simplest form

h $0.25 = \frac{\boxed{}}{\boxed{}} = \frac{\boxed{}}{\boxed{}}$
Simplest form

i $0.022 = \frac{\boxed{}}{\boxed{}} = \frac{\boxed{}}{\boxed{}}$
Simplest form

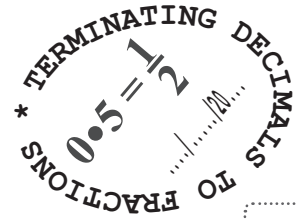
j $0.045 = \frac{\boxed{}}{\boxed{}} = \frac{\boxed{}}{\boxed{}}$
Simplest form

k $0.0028 = \frac{\boxed{}}{\boxed{}} = \frac{\boxed{}}{\boxed{}}$
Simplest form

l $0.0605 = \frac{\boxed{}}{\boxed{}} = \frac{\boxed{}}{\boxed{}}$
Simplest form



Terminating decimals to fractions



3 Write each of these decimals as equivalent mixed numerals:

a $2.3 = \begin{array}{|c|} \hline \\ \hline \\ \hline \end{array}$

b 1.1 = 

c $3.07 =$

d $1.03 =$

e 4.001 =

f $2.009 =$

4 Write each of these decimals as equivalent mixed numerals and then simplify:

a $2.8 = \boxed{}$

b $1.4 = \boxed{}$

c $4.06 = \frac{\boxed{}}{\boxed{}}$

$$= \text{[dashed rectangle]} \text{---} \text{[dashed line with crossbar]}$$

[illegible]

$$= \frac{\boxed{}}{\boxed{}}$$

Simplest form

Simplest form

Simplest form

d $3.05 =$

e $2.75 =$

f $5.005 = \frac{\boxed{}}{\boxed{}}$

$$=$$

$$= \frac{\square}{\square}$$

Simplest form

Simplest form

Simplest form

g $1.004 =$

h $2.025 = \boxed{} \div \boxed{}$

i $3.144 = \frac{\boxed{}}{\boxed{}}$

$$= \frac{\boxed{}}{\boxed{}}$$

Simplest form

Simplest form

Simplest form

Fractions to terminating decimals

Where possible, just write as an equivalent fraction with a power of 10 in the denominator first.

numerator
denominator



$$\begin{aligned}\frac{3}{5} &= \frac{3 \times 2}{5 \times 2} \\ &= \frac{6}{10} \\ \therefore &= 0.6\end{aligned}$$

Multiply numerator and denominator by the same value

Equivalent fraction with a power of 10 in the denominator

Three fifths = six tenths = zero point six

Sometimes it is easier to first simplify the fraction before changing to a decimal.

Write these as an equivalent decimal

(i) $\frac{3}{12}$

$$\frac{3 \div 3}{12 \div 3} = \frac{1}{4}$$

Simplify fraction

$$\frac{1}{4} = \frac{1 \times 25}{4 \times 25}$$

$$= \frac{25}{100}$$

Equivalent fraction with a power of 10 in the denominator

$$\therefore = 0.25$$

Three twelfths = one quarter = twenty five hundredths = zero point two five

(ii) $2\frac{3}{15}$

$$2\frac{3 \div 3}{15 \div 3} = 2\frac{1}{5}$$

Simplify fraction part

$$2\frac{1 \times 2}{5 \times 2} = 2\frac{2}{10}$$

$$= 2.2$$

Equivalent fraction with a power of 10 in the denominator

Two and three fifteenths = two and one fifth = two and two tenths = two point two



Fractions to terminating decimals

include a leading zero



1 Write each of these fractions as equivalent decimals.

a $\frac{9}{10} = \boxed{}$ b $\frac{3}{100} = \boxed{}$ c $\frac{11}{100} = \boxed{}$ d $\frac{7}{1000} = \boxed{}$

2 Write each of these as equivalent fractions with a power of 10 in the denominator.

a $\frac{1}{2} = \frac{\boxed{}}{\boxed{}}$ b $\frac{2}{5} = \frac{\boxed{}}{\boxed{}}$ c $\frac{3}{4} = \frac{\boxed{}}{\boxed{}}$ d $\frac{9}{20} = \frac{\boxed{}}{\boxed{}}$

e $\frac{8}{25} = \frac{\boxed{}}{\boxed{}}$ f $\frac{3}{250} = \frac{\boxed{}}{\boxed{}}$ g $\frac{11}{200} = \frac{\boxed{}}{\boxed{}}$ h $\frac{2}{125} = \frac{\boxed{}}{\boxed{}}$

i $1\frac{4}{5} = \boxed{}\frac{\boxed{}}{\boxed{}}$ j $3\frac{1}{25} = \boxed{}\frac{\boxed{}}{\boxed{}}$ k $6\frac{7}{20} = \boxed{}\frac{\boxed{}}{\boxed{}}$

3 (i) Write each of these as equivalent fractions with a power of 10 in the denominator.
(ii) Change to equivalent decimals.

a $\frac{1}{5} = \frac{\boxed{}}{\boxed{}} = \boxed{}$ b $\frac{1}{4} = \frac{\boxed{}}{\boxed{}} = \boxed{}$ c $\frac{11}{25} = \frac{\boxed{}}{\boxed{}} = \boxed{}$

d $\frac{4}{25} = \frac{\boxed{}}{\boxed{}} = \boxed{}$ e $\frac{1}{200} = \frac{\boxed{}}{\boxed{}} = \boxed{}$ f $\frac{6}{125} = \frac{\boxed{}}{\boxed{}} = \boxed{}$

g $2\frac{9}{25} = \boxed{}\frac{\boxed{}}{\boxed{}} = \boxed{}$ h $1\frac{1}{200} = \boxed{}\frac{\boxed{}}{\boxed{}} = \boxed{}$ i $8\frac{7}{50} = \boxed{}\frac{\boxed{}}{\boxed{}} = \boxed{}$

**Fractions to terminating decimals**

4 Change each of these fractions to equivalent decimals after first simplifying. Show **all** your working.

a $\frac{12}{20}$

b $\frac{20}{25}$

c $\frac{18}{24}$

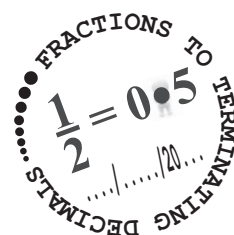
d $\frac{22}{40}$

e $\frac{9}{75}$

f $3\frac{12}{40}$

g $1\frac{36}{600}$

h $2\frac{12}{150}$





Fractions to terminating decimals

When changing the denominator to a power of 10 is not easy, you can write the numerator as a decimal and then divide it by the denominator.

Write this fraction as an equivalent decimal



If you need more decimal place 0s, you can add them in later!

$$\frac{5}{8} = 5.000 \div 8$$

Write numerator as a decimal and divide by the denominator

$$= 8 \overline{) 5.000}$$

$$= 8 \overline{) 5.625}$$

Complete division, keeping the decimal point in the same place

$$\therefore = 0.625$$

Five eighths = zero point six two five

5 Complete these divisions to find the equivalent decimal:

a $\frac{2}{5} = 2.000 \div 5$

$$= 5 \overline{) 2.000}$$

=

b $\frac{1}{4} = 1.000 \div 4$

$$= 4 \overline{) 1.000}$$

=

c $\frac{3}{8} = 3.000 \div 8$

$$= 8 \overline{) 3.000}$$

=

d $\frac{8}{5} = 8.000 \div 5$

$$= 5 \overline{) 8.000}$$

=

e $\frac{11}{8} = 11.000 \div 8$

$$= 8 \overline{) 11.000}$$

=

f $\frac{27}{4} = 27.000 \div 4$

$$= 4 \overline{) 27.000}$$

=

**Fractions to terminating decimals**

- 6 Simplify these fractions and then write as an equivalent decimal using the division method. Show **all** your working.

a $\frac{12}{15}$

b $\frac{9}{12}$

c $\frac{49}{56}$

d $\frac{18}{8}$

e $\frac{81}{24}$

f $\frac{26}{16}$



Adding and subtracting decimals

Just add or subtract the digits in the same place value.

To do this, line up the decimal points and matching place values vertically first.

- Add 2.45 to 6.31 (i.e. $2.45 + 6.31$)

$$\begin{array}{r} 2.45 + \\ 6.31 \\ \hline 8.76 \end{array}$$

Decimal points lined up vertically

Add matching place values together

- Subtract 5.18 from 11.89 (i.e. $11.89 - 5.18$)

$$\begin{array}{r} 11.89 - \\ 5.18 \\ \hline 6.71 \end{array}$$

Decimal points lined up vertically

Subtract matching place values

Calculate each of these further additions and subtractions

- (i) $24.105 + 11.06 + 6.5902$

$$\begin{array}{r} 24.105 + \\ 11.06 \\ 6.5902 \\ \hline 41.7552 \end{array}$$

Decimal points lined up vertically

Add matching place values together



Any place value spaces
are treated as 0s

$$\therefore 24.105 + 11.06 + 6.5902 = 41.7552$$

Rounding decimal values before adding is sometimes used to quickly approximate the size of the answer.

- (ii) Round each value in question (i) to the nearest whole number before adding.

$$\begin{aligned} \therefore 24.105 + 11.06 + 6.5902 &\approx 24 + 11 + 7 \\ &\approx 42 \end{aligned}$$

Values rounded to nearest ones

Approximate value for addition

Note: Rounding values **before** adding/subtracting is not as accurate as rounding **after** adding/subtracting.

- (iii) $80.09 - 72.6081$

$$\begin{array}{r} 80.09 - 72.6081 \\ \hline 7.4819 \end{array}$$

Decimal points lined up vertically

Subtract matching place values

$$\therefore 80.09 - 72.6081 = 7.4819$$



Fill place value spaces
in the top number with
a '0' when subtracting



Adding and subtracting decimals

1 Complete these additions and subtractions:

a
$$\begin{array}{r} 0.14 + \\ 0.73 \\ \hline \end{array}$$

b
$$\begin{array}{r} 1.68 + \\ 5.30 \\ \hline \end{array}$$

c
$$\begin{array}{r} 0.246 + \\ 0.832 \\ \hline \end{array}$$

d
$$\begin{array}{r} 12.194 + \\ 9.057 \\ \hline \end{array}$$

e
$$\begin{array}{r} 0.99 - \\ 0.26 \\ \hline \end{array}$$

f
$$\begin{array}{r} 5.074 - \\ 1.064 \\ \hline \end{array}$$

g
$$\begin{array}{r} 5.24 - \\ 0.83 \\ \hline \end{array}$$

h
$$\begin{array}{r} 24.158 - \\ 13.694 \\ \hline \end{array}$$

2 Calculate these additions and subtractions, showing all working:

a Add 8.75 to 1.24

b Subtract 3.15 from 4.79

c Add 0.936 to 0.865

d Add 2.19, 5.6 and 0.13

e Subtract 0.9356 from 8.6012

f Add 10.206, 4.64 and 8.0159





Adding and subtracting decimals

3 a Approximate these calculations by rounding each value to the nearest whole number first.

$$(i) \quad 5.7 + 6.2 \approx \boxed{} + \boxed{} \\ \approx \boxed{}$$

$$(ii) \quad 0.9 + 9.4 \approx \boxed{} + \boxed{} \\ \approx \boxed{}$$

$$(iii) \quad 8.3 - 1.9 \approx \boxed{} - \boxed{} \\ \approx \boxed{}$$

$$(iv) \quad 11.3 - 0.2 \approx \boxed{} - \boxed{} \\ \approx \boxed{}$$

$$(v) \quad 8.34 + 1.61 + 0.54 \approx \boxed{} + \boxed{} + \boxed{} \\ \approx \boxed{}$$

$$(vi) \quad 2.71 + 3.80 + 1.92 \approx \boxed{} + \boxed{} + \boxed{} \\ \approx \boxed{}$$

b Calculate parts (v) and (vi) again, this time rounding **after** adding the numbers to get a more accurate approximate value.

$$(i) \quad 8.34 + 1.61 + 0.54$$

$$(ii) \quad 2.71 + 3.80 + 1.92$$

4 Calculate these subtractions, showing all your working:

a $7.8 - 2.56$

b $13.09 - 8.4621$

c $0.52 - 0.12532$

Multiplying with decimals

Just write the terms as whole numbers and multiply. Put the decimal point back in when finished.

The number of decimal places in the answer = the number of decimal places in the question!

① Calculate 4×1.2

$$4 \times 12 = 48$$

Multiply both terms as whole numbers

$$\begin{array}{r} 1 \\ \swarrow \\ 48 \end{array}$$

1 decimal place in question = 1 decimal place in answer

$$\therefore 4 \times 1.2 = 4.8$$

② Calculate 0.02×1.45

$$2 \times 145 = 290$$

Multiply both terms as whole numbers

$$\begin{array}{r} 4 \ 3 \ 2 \ 1 \\ \swarrow \swarrow \swarrow \swarrow \\ 290 \end{array}$$

$$\therefore 0.02 \times 1.45 = 0.0290$$

4 decimal places in question = 4 decimal places in answer

How does this work when multiplying with decimals? Excellent question! Very glad you asked!

Let's do the second one again but this time change the decimals to equivalent fractions first

$$0.02 \times 1.45 = \frac{2}{100} \times \frac{145}{100}$$

Changing the decimals to fractions

$$= \frac{2 \times 145}{100 \times 100}$$

Multiply numerators and denominators together

$$= \frac{290}{10\,000}$$

Number of zeros in denominator = total of decimal places in question

$$= 290 \div 10\,000$$

$$= \begin{array}{r} 4 \ 3 \ 2 \ 1 \\ \swarrow \swarrow \swarrow \swarrow \\ 0.290 \end{array}$$

Dividing by 10 000 moves decimal point four places to the left

$$= 0.0290$$

$\therefore$ 4 decimal places in question = 4 decimal places in answer



Try this method for yourself on the first example above, remembering that $4 = \frac{4}{1}$ as a fraction.

**Multiplying with decimals**

1 Calculate these whole number and decimal multiplications, showing all you working:

a 0.8×2

b 5×1.5

c 0.14×6

d 0.62×4

e 3×0.032

f 1.134×2



2 Calculate these decimal multiplications, showing all your working:

a 3.8×0.2

b 1.09×0.08

c 2.7×2.5

d 7.1×1.4

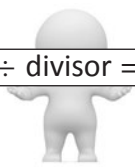
e 3.21×2.1

f 17.2×9.3

Dividing with decimals

Opposite to multiplying, we move the decimal point before dividing if needed.

To find the quotient involving decimals, the question must be changed so the divisor is a whole number.



$$\text{dividend} \div \text{divisor} = \text{quotient}$$

- Calculate $4.28 \div 4$

$$\begin{array}{r} 1.07 \\ 4 \overline{)4.28} \end{array}$$

Divisor already a whole number so no change needed

$$\therefore 4.28 \div 4 = 1.07$$

- Calculate $0.0456 \div 0.006$

$$0.0456 \div 0.006 = 0.0456 \div 0.006 \quad \text{Move both decimal points right until divisor is a whole number}$$

$$= 45.6 \div 6$$

$$\begin{array}{r} 07.6 \\ 6 \overline{)45.6} \end{array}$$

Quotient > Dividend
if divisor < 1



$$\therefore 0.0456 \div 0.006 = 7.6$$

Drop off any 0s at the front of the answer

Here's another example showing how to treat remainders

Calculate $1.26 \div 0.8$

$$1.26 \div 0.8 = 1.26 \div 0.8$$

Move both decimal points right until divisor is a whole number

$$= 12.6 \div 8$$

$$\begin{array}{r} 01.575 \\ 8 \overline{)12.600} \end{array}$$

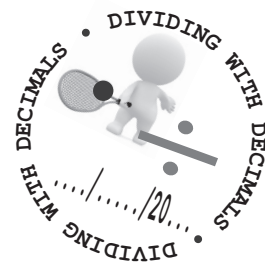
Add 0s on the end of the dividend for each new remainder

$$\therefore 1.26 \div 0.8 = 1.575$$

Drop off any 0s at the front



Dividing with decimals



1 Calculate these decimal and whole number divisions:

a $3.6 \div 4$

)

$\therefore 3.6 \div 4 =$

b $17.5 \div 5$

)

$\therefore 17.5 \div 5 =$

c $16.2 \div 9$

)

$\therefore 16.2 \div 9 =$

d $0.63 \div 3$

)

$\therefore 0.63 \div 3 =$

e $0.489 \div 5$

)

$\therefore 0.489 \div 5 =$

f $10.976 \div 7$

)

$\therefore 10.976 \div 7 =$

2 Calculate these decimal divisions, showing all your working:

a $5.2 \div 0.4$

)

$\therefore 5.2 \div 0.4 =$

b $9.6 \div 0.6$

)

$\therefore 9.6 \div 0.6 =$

c $0.56 \div 0.8$

)

$\therefore 0.56 \div 0.8 =$

d $1.58 \div 0.4$

)

$\therefore 1.58 \div 0.4$
 $=$

e $0.8125 \div 0.05$

)

$\therefore 0.8125 \div 0.05$
 $=$

f $5.3682 \div 0.006$

)

$\therefore 5.3682 \div 0.006$
 $=$

Recurring decimals

Non-terminating decimals have decimal parts that do not stop. They keep going on and on.

0.3582942049 ... Three dots means it keeps going

If the decimal parts have a repeating number pattern, they are called recurring decimals.

5.212121 ... The pattern 21 keeps repeating in the decimal parts

Here are some examples involving recurring decimals

A dot above the start and end digit of the repeating pattern is used to show it is a recurring decimal.

(i) Write these recurring decimals using the dot notation

α) 10.81818...

10.81818 ...

↑ ↑
Start End

Identify the start and end of the repeating pattern

= 10. $\dot{8}1$

Dot above start and end of the repeating pattern

β) 0.2052052...

0.2052052 ...

↑ ↑
Start End

Identify the start and end of the repeating pattern

= 0. $\dot{2}0\dot{5}$

Dot above start and end of the repeating pattern

γ) 1.047777...

1.047777 ...

↑
Start and End

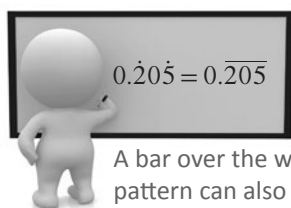
Identify the start and end of the repeating pattern

= 1.04 $\dot{7}$

Dot above start and end of the repeating pattern

or

= 1.04 $\bar{7}$



A bar over the whole pattern can also be used instead of dots

(ii) Calculate $0.1 \div 0.6$

$$0.\overset{\curvearrowright}{1} \div 0.\overset{\curvearrowright}{6} = 1 \div 6$$

$$= 1.0000 \div 6$$

Write 1 as a decimal with a few 0s

$$\begin{array}{r} 0.1666... \\ 6 \overline{) 1.0000} \\ \underline{6} \\ 40 \\ \underline{40} \\ 0 \\ \underline{0} \\ 0 \\ \underline{0} \\ 0 \end{array}$$

Repeats the same remainder when dividing

$$\therefore 1 \div 6 = 0.1666... = 0.1\dot{6}$$

Recurring decimal in simplest notation



Recurring decimals

- 1 What is the name of the horizontal line above the repeated numbers in a recurring decimal?

Highlight the boxes that match the recurring decimals in each row with the correct simplified notation in each column to find the answer.

Not all of the matches form part of the answer!

	$0.\dot{1}4$	$0.\bar{4}$	$41.\dot{1}$	$0.\overline{144}$	$0.\dot{1}4\dot{1}$	$0.4\dot{1}$	$4.\overline{14}$	$0.\dot{4}0\dot{1}$	$4.\dot{1}$	$0.\dot{4}\dot{1}$
4.1414 ...	C ϕ	F η	N δ	W γ	D β	A α	U μ	P ρ	L ϵ	O λ
0.144144 ...	Y μ	A λ	R ϵ	T ρ	K ϕ	E η	R δ	I γ	U β	S α
0.1444 ...	L α	D β	A λ	I η	M ρ	B ϵ	S γ	A δ	U ϕ	Q μ
0.401401 ...	R η	Z δ	A μ	E ϕ	A γ	N ρ	O α	M β	A η	G ϵ
4.111 ...	A ϵ	T ϕ	P γ	H δ	T α	Y μ	A ρ	A η	C λ	A β
0.4111...	I δ	Y ρ	A β	U μ	H λ	I ϕ	E ϵ	S λ	I ρ	T α
0.4141 ...	A β	L α	D ρ	E ϵ	A δ	N γ	L λ	E ϕ	O δ	N η
41.111 ...	W γ	J ϵ	B δ	A α	X η	M λ	A β	U μ	A	A ϕ
0.444 ...	P λ	V γ	E α	F β	A μ	B δ	T	Y ϵ	E γ	I ρ
0.1411411 ...	H ρ	A μ	A λ	A λ	U ϵ	A β	A η	A α	D δ	R γ

γ ϕ η λ μ α ϵ β

- 2 Calculate these divisions which have recurring decimals as a result.

Write answers using dot notation.

a $1 \div 3$

)

$\therefore 1 \div 3 =$

b $4 \div 9$

)

$\therefore 4 \div 9 =$

c $5 \div 6$

)

$\therefore 5 \div 6 =$

d $1.6 \div 6$

)

$\therefore 1.6 \div 6 =$

e $2.5 \div 9$

)

$\therefore 2.5 \div 9 =$

f $0.34 \div 3$

)

$\therefore 0.34 \div 3 =$



Recurring decimals



- 3 (i) Complete the following divisions to five decimal places.
 (ii) Determine whether the answer is a recurring decimal or not.

a $2 \div 3$

)

$\therefore 2 \div 3 =$

Recurring decimal? ☐ Yes ☐ No

b $1 \div 6$

)

$\therefore 1 \div 6 =$

Recurring decimal? ☐ Yes ☐ No

c $1 \div 7$

)

$\therefore 1 \div 7 =$

Recurring decimal? ☐ Yes ☐ No

d $1.6 \div 7$

)

$\therefore 1.6 \div 7 =$

Recurring decimal? ☐ Yes ☐ No

e $2.9 \div 3$

)

$\therefore 2.9 \div 3 =$

Recurring decimal? ☐ Yes ☐ No

f $0.33 \div 0.8$

)

$\therefore 0.33 \div 0.8 =$

Recurring decimal? ☐ Yes ☐ No

g $0.68 \div 0.3$

)

$\therefore 0.68 \div 0.3$
 $=$

Recurring decimal? ☐ Yes ☐ No

h $0.019 \div 0.06$

)

$\therefore 0.019 \div 0.06$
 $=$

Recurring decimal? ☐ Yes ☐ No

i $0.00644 \div 0.002$

)

$\therefore 0.00644 \div 0.002$
 $=$

Recurring decimal? ☐ Yes ☐ No



Simple recurring decimals into single fractions

Only recurring, non-terminating decimals can be written in fraction form.

Here is a quick way for simple decimals with the pattern starting right after the decimal point.

$$0.111... = 0.\dot{1} = \frac{1}{9}$$

One digit in repeating pattern, so that digit over 9

$$0.1212... = 0.\dot{1}2 = \frac{12}{99}$$

Two digits in repeating pattern, so those two digits over 99

$$= \frac{12 \div 3}{99 \div 3}$$

$$= \frac{4}{33}$$

Always simplify
fractions



$$0.301301... = 0.\dot{3}0\dot{1} = \frac{301}{999}$$

Three digits in repeating pattern, so those three digits over 999

Here are some other examples including mixed numerals.

Write each of these recurring decimals as mixed numerals in simplest form

(i) 3.777...

$$3.7777... = 3.\dot{7}$$

One digit in repeating decimal pattern, so that digit over 9

$$= 3\frac{7}{9}$$

Digits in front of decimal point form the whole number part

(ii) 16.345345...

$$16.345345... = 16.\dot{3}4\dot{5}$$

Three digits in repeating decimal pattern, so those digits over 999

$$= 16\frac{345}{999}$$

Digits in front of decimal point form the whole number

$$= 16\frac{345 \div 3}{999 \div 3}$$

Simplify the fraction part

$$= 16\frac{115}{333}$$



Simple recurring decimals into single fractions

1 Use the shortcut method to write each of these recurring decimals as a fraction in simplest form:

a $0.\dot{4}$

b $0.\bar{8}$

c $0.\dot{6}$

d $0.\dot{1}\dot{1}$

e $0.\dot{2}\dot{7}$

f $0.\dot{5}\dot{7}$

2 Use the shortcut method to write each of these recurring decimals as mixed numerals in simplest form.

a $1.\dot{5}$

b $2.\bar{7}$

c $4.\bar{3}$

d $3.\bar{6}$

e $5.\overline{12}$

f $0.\dot{1}\dot{1}\dot{7}$

g $0.\overline{162}$

h $5.\overline{1485}$

i $0.\dot{4}89\dot{6}$

3 (i) Write $0.\dot{9}$ as a fraction in simplest form.

(ii) Does anything unusual seem to be happening with your answer? Explain.



Combining decimal techniques to solve problems

All the techniques in this booklet can be used to solve problems.



These examples show different ways decimals pop up in every-day life

- (i) These rainfall measurements were taken during three days of rain from a small weather gauge:

13.8 mm 36.1 mm 27.6 mm

What was the total rainfall for the three days, to the nearest whole mm?

$$13.8 +$$

Add the decimal values together

$$36.1$$

$$27.6$$

$$\hline 77.5$$

$$\approx 78 \text{ mm}$$

Round to nearest whole mm

$\therefore$ The total rainfall over the three days was approximately 78 mm Answer with a statement

- (ii) The results for five runners in a 100 m race were plotted on the number line below.



- α) What was the fastest time run (to the nearest thousandth of a second)?

Fastest time = left-most plotted point = 11.221 seconds

- β) What time did two runners finish the race together on?

Two runners with the same time = two dots at the same point = 11.223 seconds

- γ) What was the average time ran by all runners in this race?

Average time = The sum of all the times ran divided by the number of runners

$$= (11.221 + 11.223 + 11.223 + 11.226 + 11.228) \div 5 \quad \text{Read off all the times}$$

$$= 56.121 \div 5$$

Add, then divide by 5

$$= \begin{array}{r} 11.2242 \\ 5 \overline{)56.1210} \end{array}$$

The average time ran by all the runners in the race = 11.2242 seconds

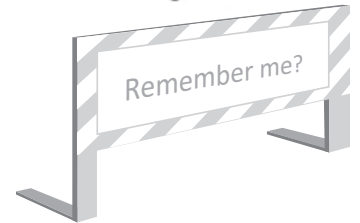
Answer with a statement



Combining decimal techniques to solve problems

- 1 To make dark-green coloured paint, you can mix yellow and blue together, using exactly 0.5 (half) as much yellow as you do blue.

- a Use multiplication to show how much yellow paint you will need if you use **all** of the 12.46 mL of blue paint you have.



- b How many millilitres of dark-green paint can you make with 18.45 mL of yellow paint in the mix? Round your answer to the nearest tenth of a mL.

- 2 Derek types his essays at an average speed of 93.45 words every minute. How many words does he type in five minutes (to the nearest whole word)?



- 3 Nine people were trying out for a speed roller skating team around an oval flat track. The shortest time to complete six full laps of the track for each person were recorded on the number line below:



- a What was the slowest time recorded to 3 decimal places?
- b To make the team, a skater had to complete the six laps in less than 126.245 seconds. How many skaters made it into the team?



- c How many skaters missed out making the team by **less** than 0.01 seconds?

**Combining decimal techniques to solve problems**

- 4 The wireless transmitter in Laura's house reduces in signal strength by 0.024 for every 1 metre of distance she moves her computer away from the transmitters antenna. Her computer displays signal strength using bars as shown below:

	4 bars = 0.81 to 1.0 signal strength
	3 bars = 0.61 to 0.8 signal strength
	2 bars = 0.41 to 0.6 signal strength
	1 bar = 0.21 to 0.4 signal strength
	0 bars = 0.2 or below signal strength



How many bars of signal strength would Laura have if using her computer 16.25m away from the antenna?

- 5 Ruofan is putting together a video of a recent karaoke party with her friends. She will be using five of her favourite music tracks for the video.

The length of time each of the tracks play for is:

3.55 min, 5.14 min, 2.27 min, 3.18 min and 4.86 min

If she uses the entire length of the tracks with a 0.15 min break in each of the four gaps between songs, how long will her video run for (to the nearest whole minute)? Show all your working.



**Combining decimal techniques to solve problems**

- 6 After a recent study by a city council, the average number of people in each household was determined to be 3.4. Explain how this is possible if a household cannot actually have 0.4 of a person?
psst: Check example on page 33 to see how average calculations are made.

- 7 A Mexican chef has split up a mystery ingredient "Sal-X" into four exactly identical quantities in separate jars. He then distributes 138.2 mL of the secret ingredient "Sa-Y" amongst the four jars, producing in total 863.9 mL of the special sauce "SalSa-XY". How much of the mystery ingredient "Sal-X" is there in each jar (to the nearest mL)? Show all your working.



- 8 After completely flat water conditions (waves with a height of 0.0m), the height of the waves at a local beach start increasing by 0.2 m every 0.3 hours. If the waves need to be at least 1.4 metres high before surfers will ride them at this beach, how long will it be until people start surfing there to the nearest minute? Show all your working.

psst: 1.0 hours = 60 minutes



**Reflection Time**

Reflecting on the work covered within this booklet:

- 1 What useful skills have you gained by learning about decimals?

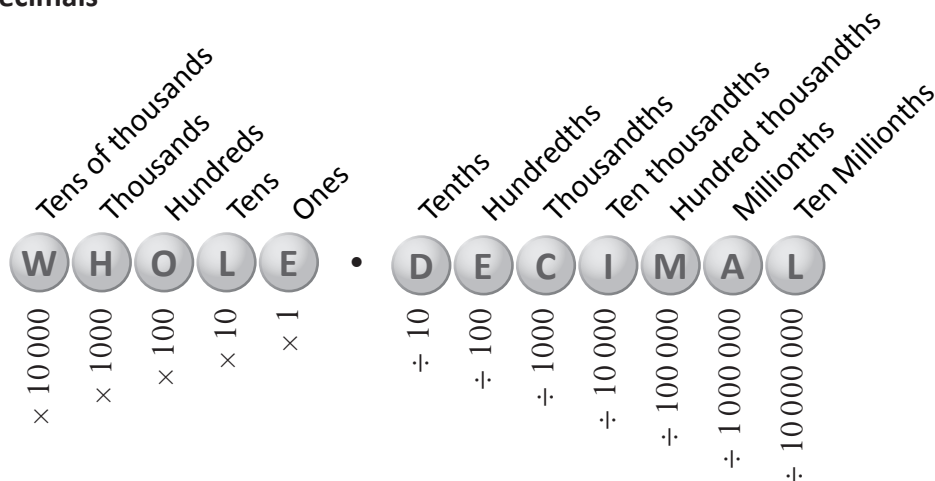
- 2 Write about one or two ways you think you could apply decimals to a real life situation.

- 3 If you discovered or learnt about any shortcuts to help with decimals or some other cool facts, jot them down here:



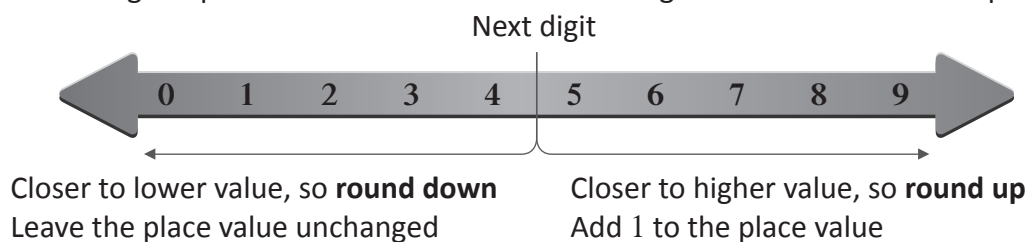
Here is a summary of the things you need to remember for decimals

Place value of decimals



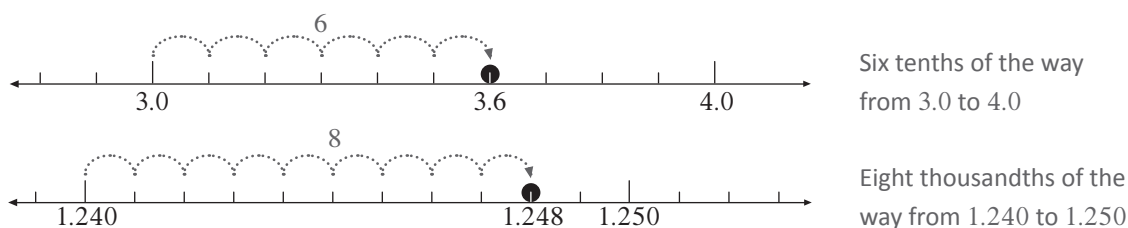
Approximations through rounding numbers

The next digit following the place value where a number is being rounded off to is the important part.



Decimals on the number line

The smallest place value in a decimal is used to position points accurately on a number line.



Multiplying and dividing by powers of ten

Move the decimal point depending on the number of zeros

$\otimes$ = decimal point moves right , $\oslash$ = decimal point moves left

$5 \times 1000 = 5.0 \times 1000$ $= 5.0$ (with arrows moving the decimal point 3 places right) $= 5000$	$8 \div 100 = 8.0 \div 100$ $= 8.0$ (with arrows moving the decimal point 2 places left) $= 0.08$
--	---

Terminating decimals to fractions

The place value of the **last digit on the right** helps us to write it as a fraction.

	Decimal	Fraction		Decimal	Fraction
Write 0.3 as a fraction:	0.3	$= \frac{3}{10}$	Write 1.07 as a fraction:	1.07	$= 1 \frac{7}{100}$
Last digit is in tenths position	↑	↑	Last digit is in hundredths position	↑	↑

Fractions to terminating decimals

Where possible, just write as an equivalent fraction with a power of 10 in the denominator first.

Eg: $\frac{3}{5} = \frac{3 \times 2}{5 \times 2}$ Multiply numerator and denominator by the same value
 $= \frac{6}{10}$ Equivalent fraction with a power of 10 in the denominator
 $\therefore = 0.6$ Three fifths = six tenths = zero point six

When this method is not easy, write the numerator as a decimal and then divide it by the denominator.

Adding and subtracting decimals

Line up the decimal points and matching place values vertically before adding or subtracting.

Multiplying and dividing decimals

Write the terms as whole numbers and multiply. Put the decimal point back in when finished.
 The number of decimal places in the answer = the number of decimal places in the question!

Eg: $\bullet 4 \times 1.2 = 4.8$ $\bullet 0.02 \times 1.45 = 0.0290$

Dividing with decimals

The question must be changed so the divisor is a whole number first.

dividend $\div$ divisor = quotient

Eg: $\bullet 13.5 \div 0.4 = 135 \div 4$ $\bullet 89.25 \div 0.003 = 89250 \div 3$

Recurring decimals

These have decimal parts with a repeating number pattern.

Eg: $\bullet 5.212121... = 5.\dot{2}1 = 5.\overline{21}$ $\bullet 0.3698698... = 0.3\dot{6}98 = 0.3\overline{698}$
 Start ↑ End Start ↑ End

Simple recurring decimals into single fractions

Always simplify fractions

Only recurring, non-terminating decimals can be written in fraction form. This is the method for simple decimals with the pattern starting right after the decimal point.

$\bullet 0.111... = 0.\dot{1} = \frac{1}{9}$

One digit in repeating pattern, so that digit over 9

$\bullet 0.1212... = 0.\dot{1}2 = \frac{12}{99} = \frac{4}{33}$

Two digits in repeating pattern, so those two digits over 99

$\bullet 8.301301... = 8.\dot{3}01 = 8\frac{301}{999}$

Three digits in repeating pattern, so those three digits over 999, Keep whole number out the front.



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